

Multilingual Document OCR Extraction Pipeline

A technical proposal for **Beyonddatagroups** — automating structured data extraction from multilingual documents at scale.

PROPOSAL · PLANNING PHASE



Problem Statement

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Multilingual Complexity

Organizations receive documents in **multiple languages** and formats, creating fragmented processing workflows.

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Manual Extraction Bottleneck

Manual text extraction is **time-consuming** and error-prone, limiting throughput and introducing inconsistencies.

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OCR Limitations

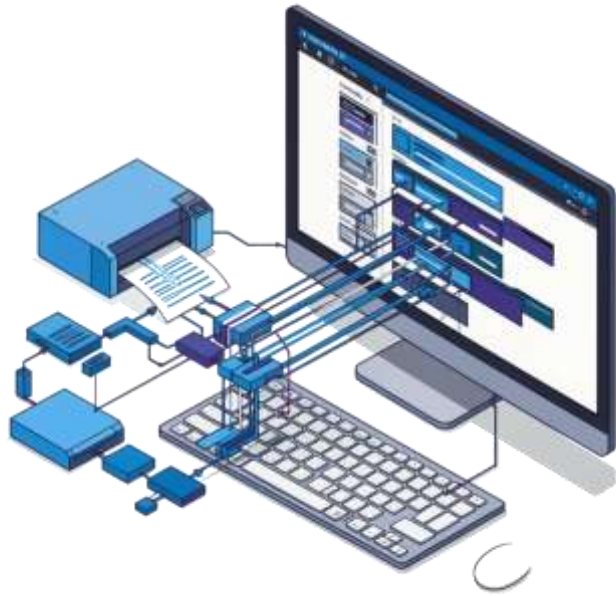
Traditional OCR performs poorly on **multilingual** documents with varied layouts and mixed scripts.

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Unstructured Output

Extracted data requires additional **cleaning and structuring** before it can be consumed by downstream systems.





EXPECTED OUTCOME

Proposed Solution

- Automated pipeline converting multilingual documents into **structured digital data**
- Support for **multiple languages** and document formats — PDF, images, scans
- High-accuracy text extraction with **layout preservation**
- Scalable, cloud-ready solution with **API integration** for downstream applications

Proposed Solution Architecture



Input Layer

Document upload via API or batch queue with validation



Processing Layer

File parsing, preprocessing, language detection, OCR, layout analysis



Output Layer

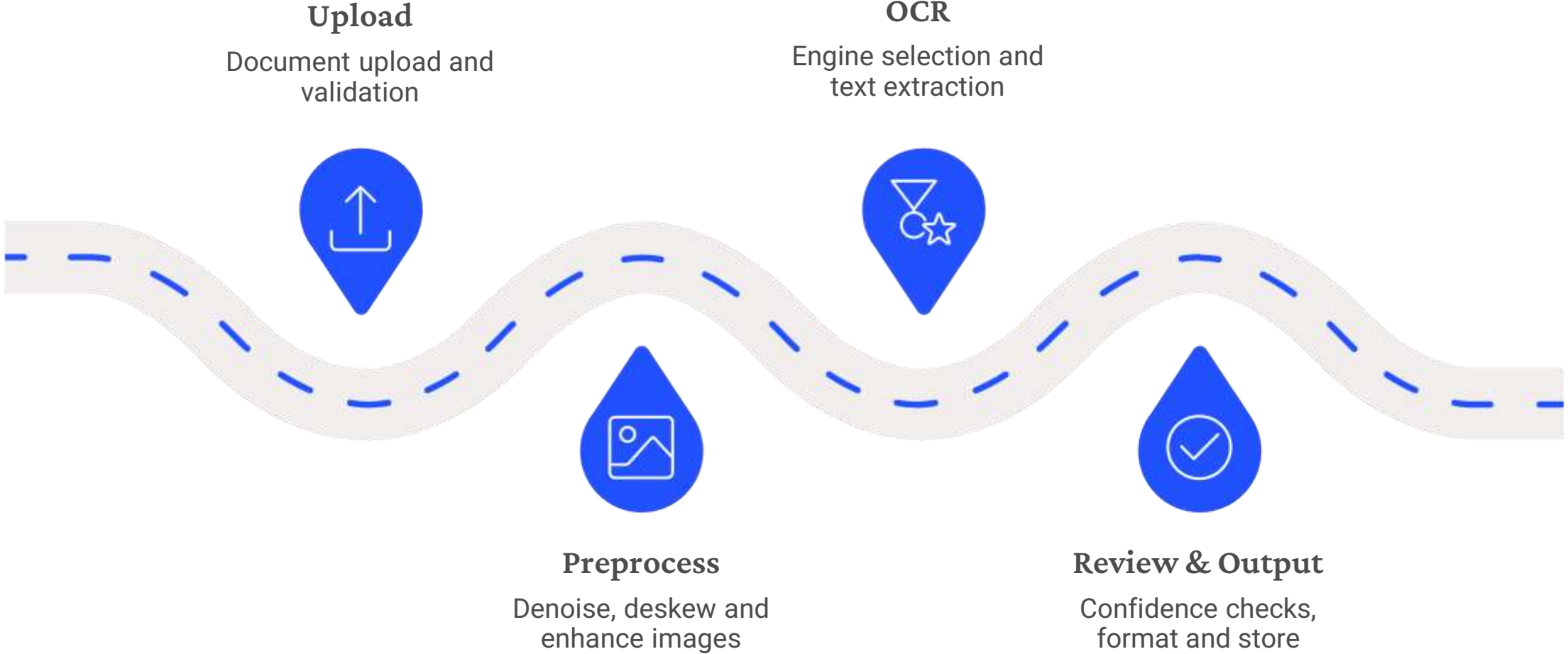
Structured JSON/TXT, database storage, downstream integration



Cross-Cutting

Logging, monitoring, error handling, and security controls

Workflow Diagram



Each stage is independently monitored — low-confidence pages are flagged for human review before final output is committed.

Technologies & Tools



Programming

Python — core language for all pipeline components



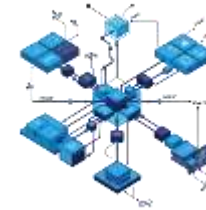
Document Processing

OpenCV, Pillow, PDF libraries for format handling and image enhancement



OCR Engine

Tesseract OCR with multilingual language packs



AI / ML

Language detection and **layout analysis models**



Backend & Storage

FastAPI/Flask, PostgreSQL, S3/GCS object storage

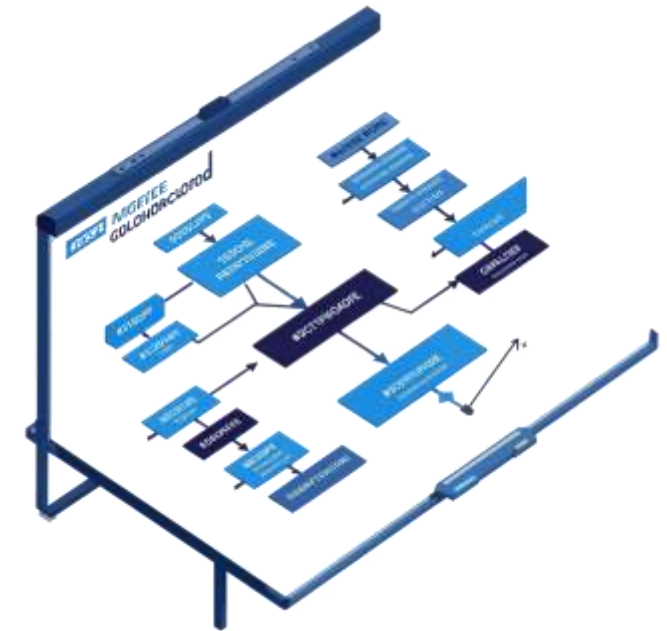


Deployment

Docker containers on AWS, GCP, or Azure cloud infrastructure

Algorithm & Approach

A deterministic, page-by-page processing algorithm ensures traceability and handles multilingual documents reliably from ingestion to structured output.



Receive & Validate

Ingest document via upload or API; validate file format (PDF, PNG, TIFF, JPG)

Preprocess

Resize, denoise, grayscale, threshold, and deskew each page image

Detect & Configure

Classify language/script; select appropriate OCR configuration per language

Extract & Output

Run OCR, clean text, validate confidence, generate structured data

Pseudocode

START

Receive and validate document

Identify file type → convert to pages/images

FOR EACH page:

Preprocess image

(resize, denoise, deskew)

Detect language / script

Select OCR configuration

Perform OCR extraction

Clean and normalize text

Check confidence threshold

IF confidence < threshold:

Flag page for human review

Combine all page results

Generate structured output (JSON / TXT)

Store result / return via API

END

- ❏ The confidence threshold check is a critical quality gate. Pages falling below the threshold are quarantined and routed to a human review queue before final data is committed.

Each page is processed independently, enabling parallel execution and granular error recovery without re-processing the entire document.

Assumptions

Supported Input Formats

Input documents are available in supported digital or scanned formats compatible with the pipeline.

Language Model Coverage

OCR language models and packs support all required target languages and scripts.

Document Quality Baseline

Document quality is sufficient for OCR extraction after standard preprocessing steps are applied.

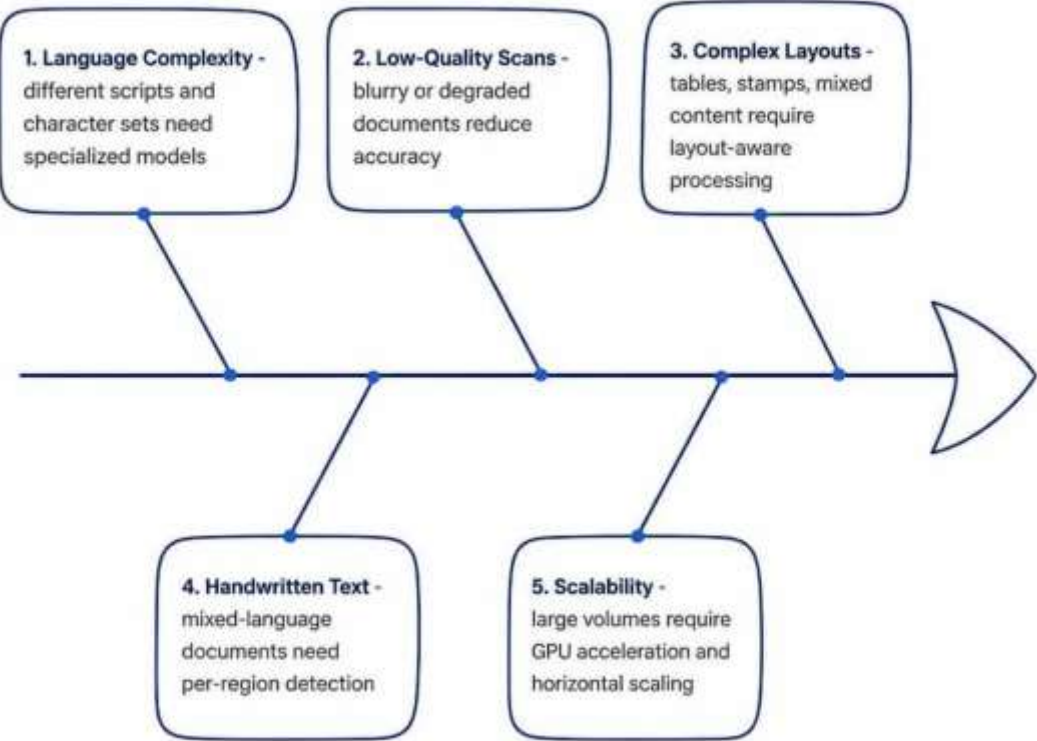
Infrastructure & Connectivity

Required compute resources, cloud/API access, and network connectivity are available for deployment.

Human Review Capacity

Human review workflows are available and staffed to handle low-confidence flagged pages.

Challenges & Risks



Language Complexity

Different scripts and character sets require specialized models tuned per language family.

Low-Quality Scans

Blurry or degraded documents significantly reduce OCR accuracy even after preprocessing.

Complex Layouts

Tables, stamps, and mixed content demand layout-aware processing beyond standard OCR.

Handwritten Text

Mixed-language documents require per-region script detection and handwriting recognition.

Scalability

High document volumes require GPU acceleration and horizontal scaling to maintain throughput.

Mitigation Strategies

1 Multilingual Models Deploy multilingual models with script-specific preprocessing pipelines per language family.	2 Advanced Denoising Implement super-resolution and advanced denoising techniques to recover low-quality scans.
3 Layout-Aware OCR Deploy layout-aware OCR engines with table reconstruction algorithms for complex documents.	4 HWR Integration Integrate dedicated Handwriting Recognition (HWR) models with per-region language detection.
5 Horizontal Scaling Design batch queues with GPU acceleration and auto-scaling cloud infrastructure for peak loads.	6 Confidence Workflows Establish confidence thresholds and structured human review queues for flagged documents.

 These mitigations should be incorporated into sprint planning from day one — not retrofitted after initial delivery.